

MRA Sentinel: Regulatory Remediation & Early Warning Proof of Concept (PoC)

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**Proactive Regulatory Tracking*

The "MRA Sentinel" is a centralized dashboard designed to migrate financial institutions from reactive, fragmented spreadsheet management to a **proactive regulatory defense**. By leveraging Natural Language Processing (NLP) and time-decay logic, the Sentinel ensures that Matters Requiring Attention (MRAs) are remediated well before they escalate into MRIs or Enforcement Actions.

The Business Case (ROI)

The Problem

- **Fragility:** MRAs managed via spreadsheets lead to "lingering status" and missed deadlines.
- **Escalation Risk:** Failure to address MRAs timely is the primary trigger for Formal Enforcement Actions and Rating downgrades (CAMELS/LFI).
- **Information Asymmetry:** Executives lack a real-time "heatmap" of regulatory risk across business lines.

The Sentinel Solution

- **Automated Ingestion:** Reduces manual entry errors by auto-mapping findings from OCC/FRB exam letters.
- **75% Early Warning System (EWS):** Triggers high-risk flags when 75% of a timeline has elapsed without status changes.
- **Audit-Ready Accountability:** Maintains a tamper-evident Audit Trail of every status change, specifically flagging "Post-Deadline" updates for internal auditors and examiners.

- **Strategic Oversight:** Heatmaps identify thematic concentrations (e.g., Cyber vs. AML) and resource bottlenecks.

Technical Acumen

- **Frontend:** [Streamlit](#) for an interactive, web-based Executive Dashboard.
- **NLP/Extraction:** [PyMuPDF](#) for document parsing and Regex-based thematic mapping.
- **Visuals:** [Altair](#) for resilient, declarative Gantt roadmaps and risk heatmaps.
- **Reporting:** [ReportLab](#) for automated generation of Executive PDF summaries.

Governance, Security & Data Privacy

Because this tool handles sensitive **Confidential Supervisory Information (CSI)**, the following protocols are recommended for production deployment:

- **Data Residency:** This PoC processes files in-memory. Production must be integrated with the bank's **Enterprise Data Lake** or a **Private Cloud** instance (AWS/Azure) that meets SOC2 standards.
- **Access Control (RBAC):** Implementation requires integration with **Single Sign-On (SSO/Active Directory)** to ensure only authorized Risk and Compliance personnel can view findings.
- **CSI Compliance:** This tool should be hosted on **internal, non-public servers**. Users must never upload live regulatory documents to public cloud instances.
- **Audit Persistence:** Production versions should write logs to a **write-once-read-many (WORM)** database to satisfy regulatory requirements for data integrity.

Deployment & Installation

1. Clone the Repository:

```
git clone https://github.com
```

2. Install Dependencies:

```
pip install -r requirements.txt
```

3. Run the Command Center:

```
streamlit run app.py
```

Audit Trail & Governance

The Sentinel implements a strict governance layer:

- **Stale Warnings:** Items with no activity for 30+ days are visually "iced" (🧊) to prompt immediate review.
- **Chronology Guard:** Prevents date inversions from corrupting roadmap visualizations.
- **Audit Context:** Automatically calculates "Days Overdue" at the moment of status transition.

Disclaimer: This tool is a Proof of Concept (PoC) only and should not be integrated with your institution's internal single-sign-on (SSO) and secure database protocols for production use.



Screenshots

Clear Files

Reset Tracker

MRA Sentinel: Command Center

Upload PDFs

FRB_Cre...sk_MRA.pdf20.1KB

FRB_MRA_TECH.pdf19.9KB

OCC_AML_MRA.pdf20.2KB

OCC_BSA_MRA.pdf20.4KB

test_OCC_letter.pdf1.1KB

+

Ingest

Executive Dashboard

Centralized Ledger

Strategic Roadmap

Alerts

Audit Trail

Executive Risk Oversight

Total MRAs

5

Max Days Overdue

484

Portfolio Health Distribution

OVERDUE

CRITICAL

WARNING

On Track

Closed

Risk Concentration Heatmap

LOB Pending

1

1

1

2

OVERDUE

CRITICAL

WARNING

On Track

Risk Tier

Clear Files

Reset Tracker

MRA Sentinel: Command Center

Upload PDFs

FRB_Cre...sk_MRA.pdf20.1KB

FRB_MRA_TECH.pdf19.9KB

OCC_AML_MRA.pdf20.2KB

OCC_BSA_MRA.pdf20.4KB

test_OCC_letter.pdf1.1KB

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Ingest

Executive Dashboard

Centralized Ledger

Strategic Roadmap

Alerts

Audit Trail

MRA_ID	Theme	Reg_Reference	Owner	Start_Date	Deadline	Status	Last_Updated	Days_Since_Update	Days_Remaining	Risk_Status
FRB-FRB_C-01	General / Other	N/A	LOB Pending	2026-03-30 00:00:00	2026-05-01 00:00:00	In Progress	2026-04-29 00:00:00	0	2	CRITICAL
FRB-FRB_M-01	Cybersecurity/IT	OCC 2013-29	LOB Pending	2026-03-30 00:00:00	2026-05-10 00:00:00	In Progress	2026-04-29 00:00:00	0	11	WARNING
OCC-OCC_A-01	Financial Crime/AML	FFIEC BSA/AML	LOB Pending	2026-03-30 00:00:00	2026-10-31 00:00:00	In Progress	2026-04-29 00:00:00	0	185	On Track
OCC-OCC_B-01	Financial Crime/AML	FFIEC BSA/AML	LOB Pending	2026-03-30 00:00:00	2027-01-31 00:00:00	In Progress	2026-04-29 00:00:00	0	277	On Track
OCC-TEST_01	Model Risk	SR 11-7	LOB Pending	2026-03-30 00:00:00	2024-12-31 00:00:00	In Progress	2026-04-29 00:00:00	0	-484	OVERDUE

Export Master Tracker

Clear Files

Reset Tracker

MRA Sentinel: Command Center

Upload PDFs

FRB_Cre...sk_MRA.pdf20.1KB

FRB_MRA_TECH.pdf19.9KB

OCC_AML_MRA.pdf20.2KB

OCC_BSA_MRA.pdf20.4KB

test_OCC_letter.pdf1.1KB

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Ingest

Executive DashboardCentralized LedgerStrategic RoadmapAlertsAudit Trail

FRB-FRB_C-01

FRB-FRB_M-01

OCC-OCC_A-01

OCC-OCC_B-01

OCC-TEST_01

2025

March

May

July

September

November

2026

March

May

July

September

November

2027

Start_Date, Deadline

Chart_Risk

OVERDUE

CRITICAL

WARNING

On Track

Closed

Export Master Tracker

Clear Files

Reset Tracker

MRA Sentinel: Command Center

Upload PDFs

FRB_Cre...sk_MRA.pdf20.1KB

FRB_MRA_TECH.pdf19.9KB

OCC_AML_MRA.pdf20.2KB

OCC_BSA_MRA.pdf20.4KB

test_OCC_letter.pdf1.1KB

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Ingest

Executive DashboardCentralized LedgerStrategic RoadmapAlertsAudit Trail

Select MRA for Alert:

FRB-FRB_C-01

Email Draft

Subject: URGENT: FRB-FRB_C-01 Alert

Dear LOB Pending,

Finding FRB-FRB_C-01 is flagged as CRITICAL.

Export Master Tracker

Clear Files

Reset Tracker

MRA Sentinel: Command Center

Upload PDFs

FRB_Cre...sk_MRA.pdf20.1KB

FRB_MRA_TECH.pdf19.9KB

OCC_AML_MRA.pdf20.2KB

OCC_BSA_MRA.pdf20.4KB

test_OCC_letter.pdf1.1KB

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Ingest

Executive DashboardCentralized LedgerStrategic RoadmapAlertsAudit Trail

	Timestamp	MRA_ID	Event	Prev	New	Audit_Context
empty						

Export Audit Log

Export Master Tracker